Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Reference frame

Meaning in my words How would you explain reference frame to a friend?	**How can you tell?** You report a cart at +5 m. What does a partner need to locate it?
Example or sketch How could a seated passenger be at rest in one frame and moving in another?	**Non-example and why** Two people report different velocities. Does that alone prove one measured incorrectly?

Components

Meaning in my words What are the components of an arrow?	**How can you tell?** If east and north are positive, which component is negative for a southeast trip?
Example or sketch What two arrows could you use to build a displacement of 3 m east and 4 m north?	**Non-example and why** A classmate counts the diagonal and both parts as three separate trips. What is counted twice?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Units

Meaning in my words What do units tell you about a number?	**How can you tell?** Before adding 2 m and 30 cm, what should you do?
Example or sketch How would you add those two lengths in metres?	**Non-example and why** A partner adds the numbers and gets 32 m. What went wrong?

Limit

Meaning in my words How would you explain limit to a friend?	**How can you tell?** Which change would make a steady-velocity forecast unreliable?
Example or sketch What limit would you write next to a one-minute straight-line prediction?	**Non-example and why** A partner uses the one-minute model for a year. What evidence would they need first?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch the reference frame and one sideways component. Box your time unit and one model assumption.



My marks mean

Two reports. One sighting.

21

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Compare position reports that use different zeros.

CHOOSE ONE JOB TOGETHER

SURVEY ENGINEER ☐

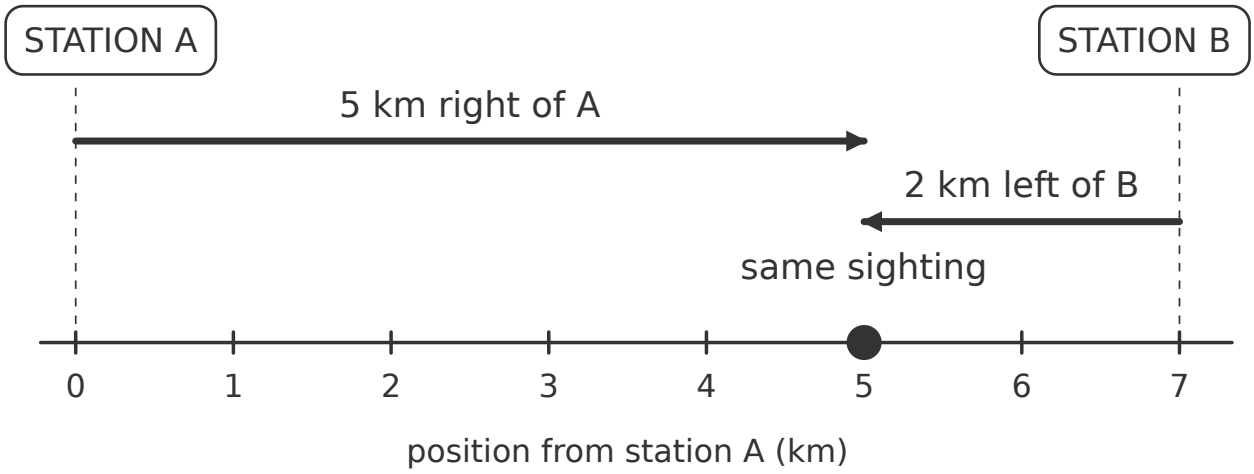
Use one axis for both stations. Give dispatch the position from A, then name what a future forecast still needs.

MAP BROADCASTER ☐

Show both stations and zeros on the locator map. Explain whether the two reports describe different locations.

Point to a clue. Keep your first marks; add the update using the color key.

Two observatories file the same sighting: "**5 km** right of station A" and "**2 km** left of station B." B is **7 km** right of A.



01 OUR FIRST CALL

Do the reports agree? Give one location and explain how the map supports it.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What number changes in the new frame? What physical location stays the same?

Switch speakers. What measurement uncertainty should accompany a future central position forecast? Give a concrete example.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks